Thermal effects and energy budgets

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Estimate a temperature change only after stating how much energy enters the material.

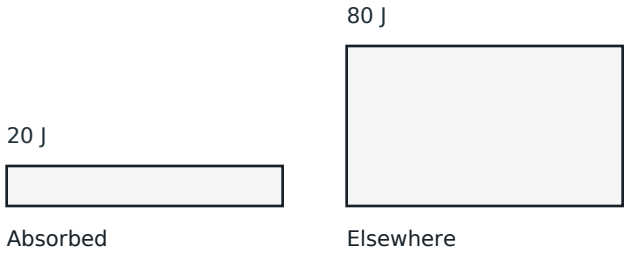
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 8.*
Work input enables the transfer; it does not contradict the direction of spontaneous heat flow.

02. Worked example: explain the model
Absorbed 250 J heats the surface under the stated model; reflected/transmitted portions go elsewhere.

03. Read and interpret the model
20 J absorbed; 80 J elsewhere.

Model / diagram check



04. Apply the physics
2°C.

05. Explain your reasoning
Phase-change latent heat, changed properties, energy transfer/losses; constant-c one-phase model no longer adequate.

06. Review: connect an earlier idea *Review: Unit 4, Day 21.*
18 J other; fraction 0.70 or 70%.

07. Exit check
Doubles.

08. STRETCH: test the model
Larger mass has half the rise for the same absorbed heat and c, absent phase change/losses.

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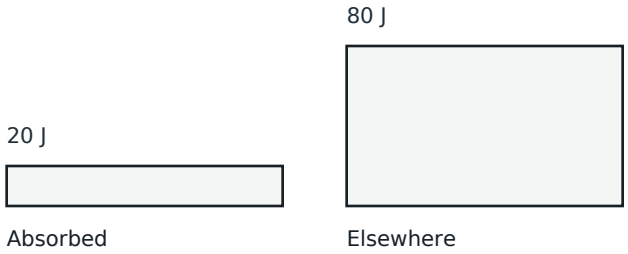
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 8.*
Total energy is conserved. At common temperature the initial temperature-difference opportunity for work is eliminated.

02. Worked example: explain the model
Absorbed 250 J heats the surface under the stated model; reflected/transmitted portions go elsewhere.

03. Read and interpret the model
20 J absorbed; 80 J elsewhere.

Model / diagram check



04. Apply the physics
Absorbed 20,000 J;ΔT 50°C.

05. Explain your reasoning
Phase-change latent heat, changed properties, energy transfer/losses; constant-c one-phase model no longer adequate.

06. Review: connect an earlier idea *Review: Unit 4, Day 21.*
Initial U=240 J; K=232 J; v≈10.77 m/s.

07. Exit check
8000 J; 200,000 J.

08. STRETCH: test the model
ΔU=900 J; ΔT=900/180=5 K.